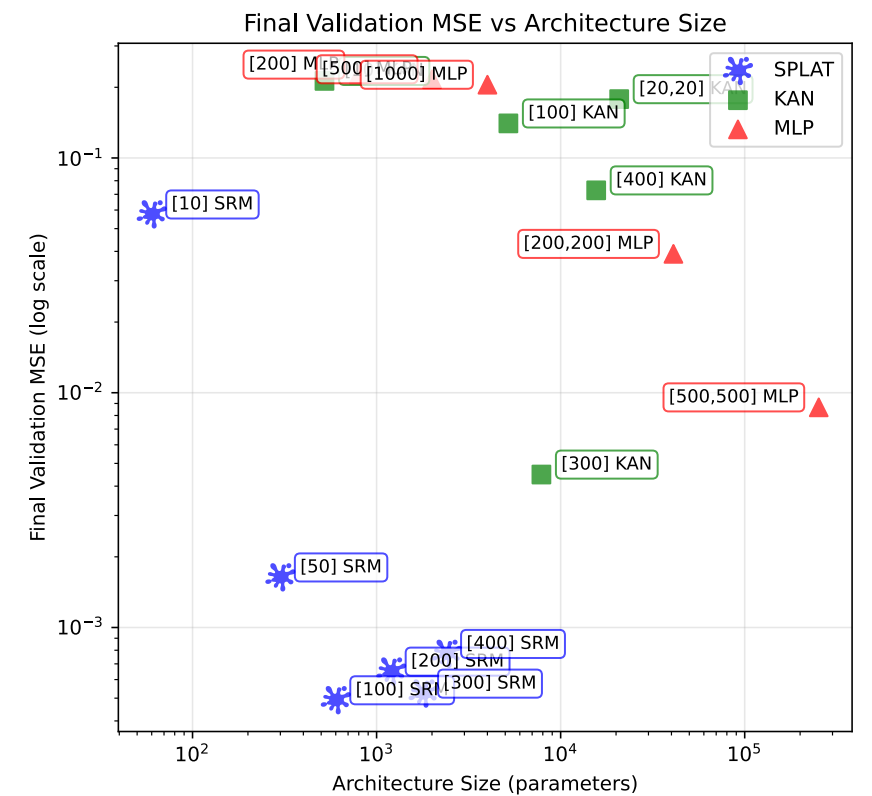
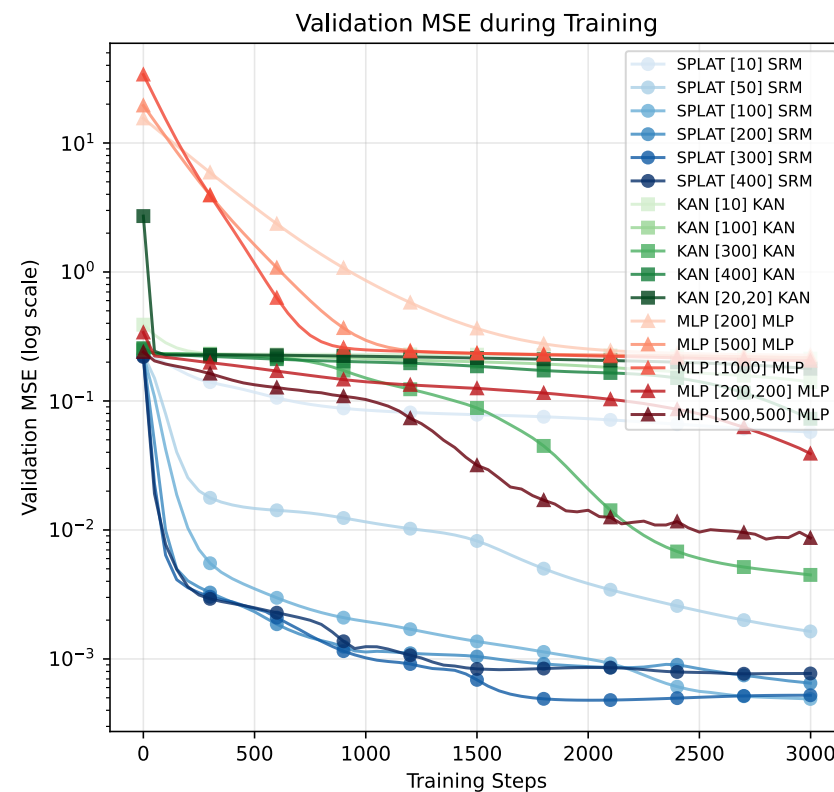
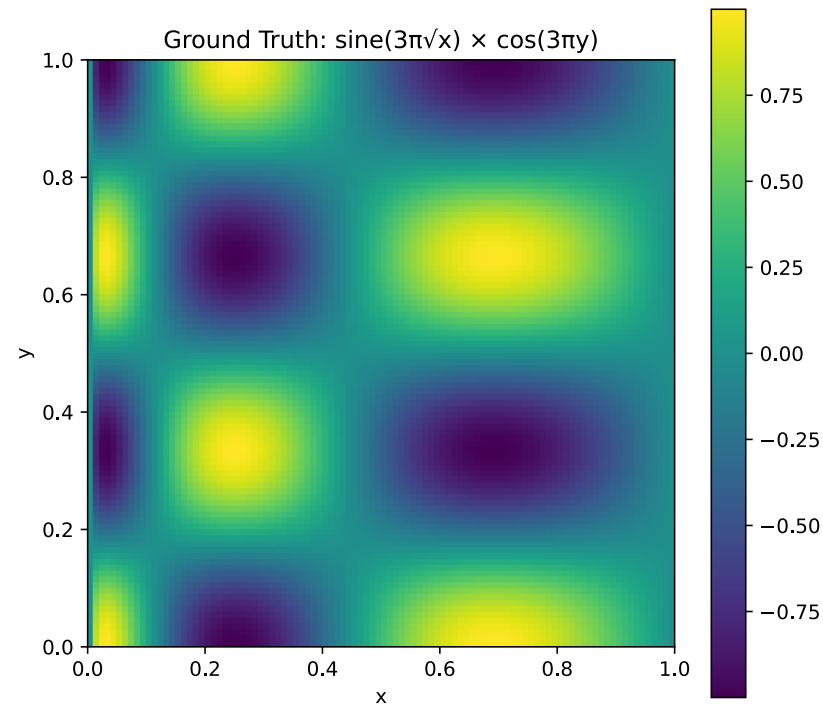
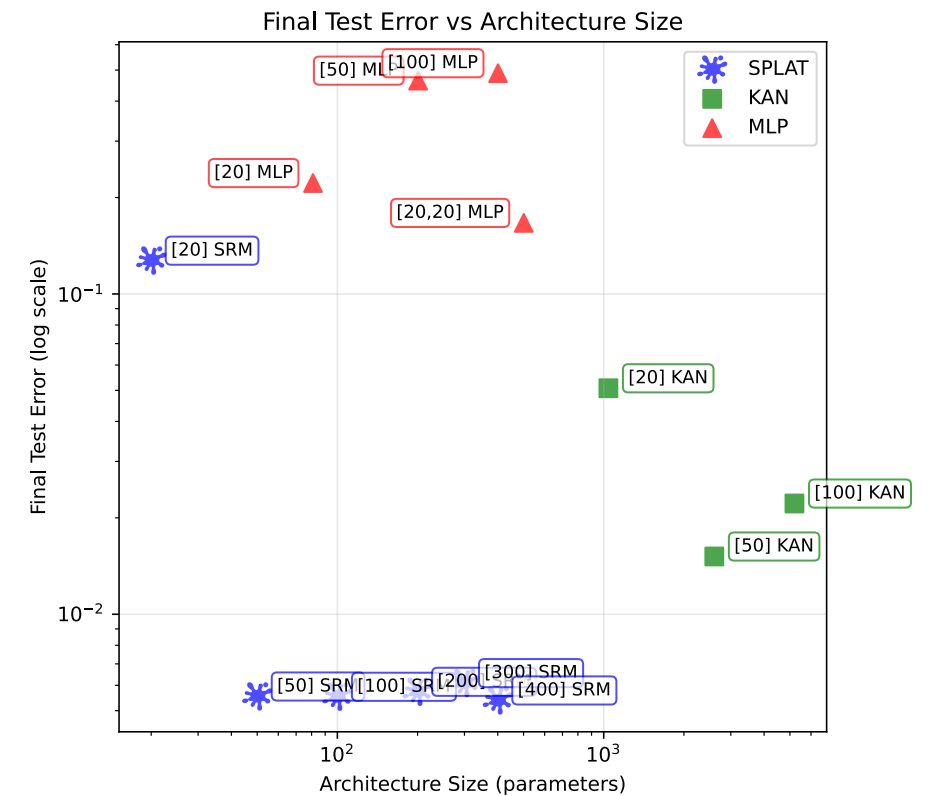
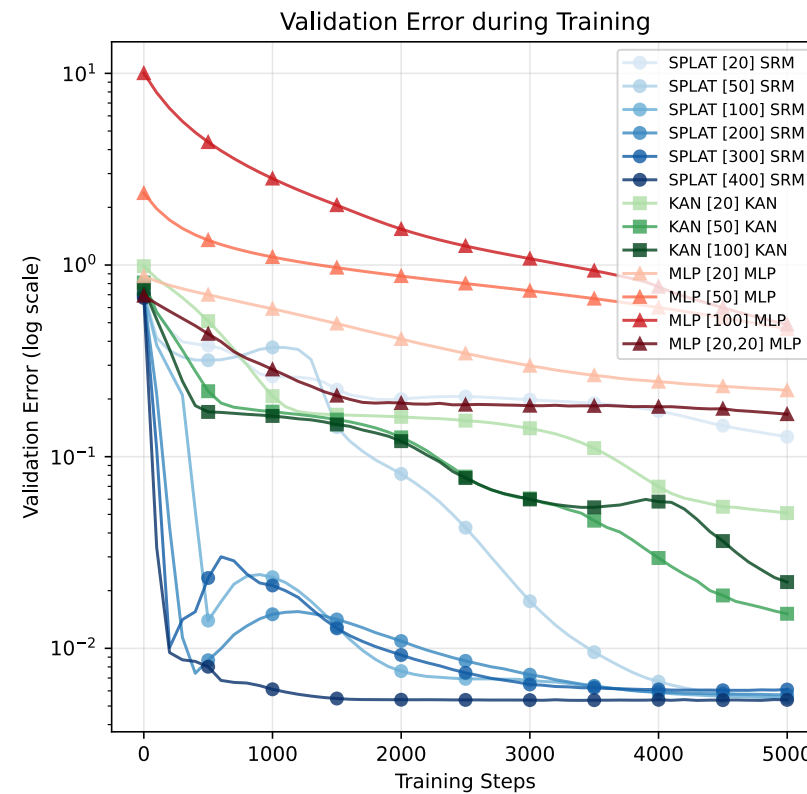
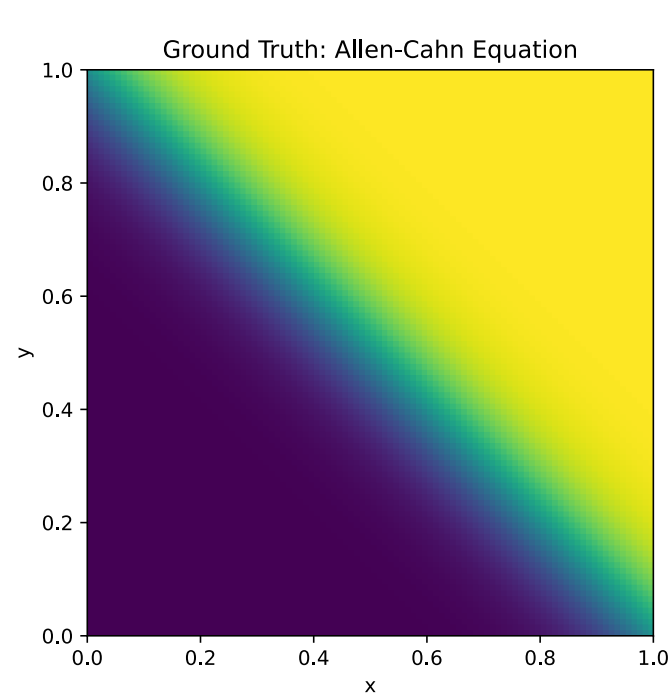


Experiments



Least squares regression: $y_i = \sin(3\pi\sqrt{x_{i,0}})\cos(3\pi x_{i,1}) + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0,0.01)$,
 $i = 1, \dots, 1000$, and $x_i \sim \mathcal{U}([0,1]^2)$ independently.

Experiments



PDE Fitting: $x_i \sim \mathcal{U}([0,1]^d)$, $i = 1 \dots 10^5$ and $z_j \sim \mathcal{U}(\partial[0,1]^d)$, $j = 1 \dots 5 \cdot 10^4$.

$$\mathcal{L}(u_\theta) = \frac{1}{n_{\text{int}}} \sum_{i=1}^{n_{\text{int}}} \|\mathcal{D}u_\theta(x_i) - f(x_i)\|_2^2 + \frac{1}{n_{\text{bdy}}} \sum_{j=1}^{n_{\text{bdy}}} \|u_\theta(z_j) - u^*(z_j)\|_2^2$$

$$\mathcal{D}u = 0.1\Delta u + u(1 - u^2)$$